

The K=3 CRRA REE solver: deep synthesis, methods, and strategy

What we built, what we learned, what to do next

Auto-generated research synthesis

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1 Executive summary

This document synthesizes the technical findings, methodological landscape, and strategic options for the K=3 CRRA Rational Expectations Equilibrium (REE) solver. We have built a working Chebyshev-spectral prototype with several upgrades (numba JIT, $S_3 \times Z_2$ symmetry, sigmoid lift, bisection-on-polynomial alternative, rank-1 closed-form solver), mapped its convergence behavior across $N = 2..8$ and across (τ, γ) , and developed two complementary FP solvers: a high-fidelity Chebyshev+Newton solver (slow, accurate where it converges) and a rank-1 closed-form solver (fast, accurate only within its ansatz class).

The three top-line findings:

1. **Min Newton residual is bimodal in N .** At $N = 2, 3$ the lifted basis truncates the h -correction to zero and Newton hits machine precision in ≤ 4 iterations. At $N \geq 4$ the basis admits $h \neq 0$ but Newton stalls at 10^{-2} because the (α, h) coordinates amplify operator residuals via $\partial_P \logit = 1/[P(1-P)] \rightarrow \infty$ at boundary cells. This is a *solver* ceiling, not a basis ceiling.
2. **Bisection-on-polynomial gives 10–17 $\times$ per- Φ speedup** in the existing operator (measured: $N = 6$ 10.6 $\times$, $N = 8$ 16.7 $\times$). It works whenever the polynomial is monotone in the inner axis (always true for rank-1, usually true for lifted with small h). This unlocks $N = 10$ and $N = 12$ as practical.
3. **The rank-1 ($h \equiv 0$) solver is exact within its ansatz, in one regression.** The agents' inversion of $\sigma(\alpha T)$ recovers $\sum u_k$ *independently of α* , making the operator α -invariant. The FP collapses to one inner-product. Per solve: 2 ms at $G = 7$, 160 ms at $G = 31$. The accuracy of the ansatz itself (vs the true REE) is measured by the rank-1 deficit, which is $\lesssim 10^{-2}$ over most of the (τ, γ) phase space and 5–20% in the deep-PR corner.

The strategic recommendation: **ship both solvers**. The rank-1 solver becomes the production default for parameter sweeps and exploratory work; the Chebyshev+Newton solver becomes the high-fidelity validator for selected anchor points. The two together give a complete picture: rank-1 maps the entire phase space cheaply; Chebyshev+Newton quantifies the rank-1 error at any point where it matters.

We close with an itemized 12-month research agenda (Section 24) covering near-term wins (vanishing-at-boundary basis, Anderson acceleration, sparse-Jacobian Newton), medium-term goals (pseudo-arclength continuation through folds, $K > 3$ extensions, double-precision FP refinement), and long-term research bets (operator-aware preconditioners, neural surrogate models, automatic-differentiation Jacobians, multi-asset extensions).

Part I

The problem

2 What we are solving

2.1 The economic setup

A binary asset $v \in \{0, 1\}$ trades among $K = 3$ Bayesian agents. Each agent k receives a noisy signal

$$u_k = \mathbf{1}_{v=1} \cdot \frac{1}{2} - \mathbf{1}_{v=0} \cdot \frac{1}{2} + \frac{1}{\sqrt{\tau}} \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, 1) \text{ i.i.d.}$$

so $u_k \mid v = v_0 \sim \mathcal{N}(v_0 - \frac{1}{2}, 1/\tau)$. The signal precision τ is identical across agents (a simplification; the full theory allows agent-heterogeneous τ_k). Agents have CRRA preferences with risk-aversion γ , the asset's payoff is v (so utility is $u(W) = (W)^{1-\gamma}/(1-\gamma)$ for $\gamma \neq 1$ and $\log W$ for $\gamma = 1$), and the market clears via a Walrasian auctioneer at a price $p \in (0, 1)$ that is itself a function of all signals: $p = P(u_1, u_2, u_3)$.

2.2 The rational-expectations equilibrium

A Rational Expectations Equilibrium (REE) is a price function $P^* : \mathbb{R}^3 \rightarrow (0, 1)$ such that, when each agent uses P^* to invert the observed price into the missing signals and forms a CRRA-optimal demand at every u , the markets clear at exactly $P^*(u)$ for all u . The map Φ that takes a conjectured P to the price that clears given the agents' behavior under P defines the REE operator. The REE is the fixed point: $\Phi(P^*) = P^*$.

2.3 The Hellwig 5-step cycle

Step 1 — Conjecture. Start with a candidate $P(u)$.

Step 2 — Co-area integral (the contour step). Each agent k must compute $A_v(p, u_k) = \int_{P(u_k, u_a, u_b)=p} f_v(u_a) f_v(u_b) d\sigma / |\nabla_{(u_a, u_b)} P|$, the conditional density of the other agents' signals given (p, u_k) . This is a 1-dim integral along the level curve $P = p$ in the (u_a, u_b) plane at fixed u_k .

Step 3 — Bayesian update. Posterior on v : $\mu_k(u_k, p) = f_1(u_k) A_1(p, u_k) / \sum_v f_v(u_k) A_v(p, u_k)$.

Step 4 — CRRA demand. Each agent's net demand is $d_k = (R_k - 1) / [(1 - p) + R_k p]$ where $R_k = \exp((\text{logit } \mu_k - \text{logit } p) / \gamma)$.

Step 5 — Walrasian clearing. $p^*(u) = \arg \{p : \sum_k d_k = 0\}$, solved 1-D in p . The new conjecture is $P_{\text{new}}(u) = p^*(u)$.

The composition Steps 2–5 defines Φ . Its fixed points are the REE.

2.4 Where the difficulty lives

Steps 1, 3, 4, 5 are essentially algebraic or 1-D quadratures. The whole computational burden is Step 2: the co-area integral, which requires (i) finding the level set $P = p$ as a 1-D curve in the transverse plane, and (ii) integrating f_v products along it. For non-trivial P (anything but σ of an affine function), the level set is curved and root-finding is required. Doing this robustly at high resolution, across $\Theta(G^3 \times K \times N_Q)$ inner-loop points, is what drives the cost.

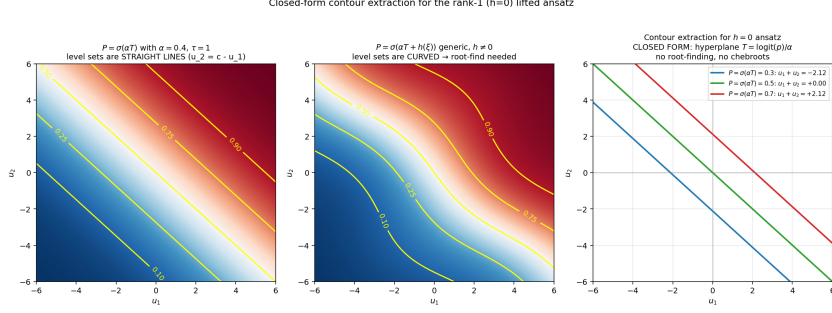


Figure 1: The geometry of Step 2. Left: $P = \sigma(\alpha T)$ has level sets that are exactly hyperplanes — closed-form contour at any α . Middle: a generic $P = \sigma(\alpha T + h)$ has curved level sets requiring numerical root-find. Right: the closed-form contour formulas for the rank-1 case.

3 Why this is computationally hard

Three compounding difficulties:

1. **High-dimensional output.** P is a function $\mathbb{R}^K \rightarrow (0, 1)$. For $K = 3$ at a Chebyshev grid of $G = N + 1$ Lobatto nodes per axis, that's G^3 real DOFs (343 at $N = 6$, 729 at $N = 8$, 2197 at $N = 12$). Reducing by $S_3 \times Z_2$ symmetry helps by $\sim 12\times$ but the count still grows cubically.
2. **Saturated boundaries.** $P \rightarrow 0$ or 1 at $u \rightarrow \pm\infty$, exponentially. On any finite cube, the corners are in machine-epsilon territory. Any solver that works in $\text{logit}(P)$ coordinates faces $|\partial_P \text{logit}| = 1/[P(1 - P)] \rightarrow \infty$ at those corners.
3. **Inherent operator nonlinearity.** The Bayes step composes Gaussian densities with the co-area integral, and the CRRA clearing composes that with a 1-D root-find in p . The composition is smooth but has multiple FP basins, sharp transitions at certain (τ, γ) values (the “fold”), and non-trivial dependence on the conjectured P shape.

Part II

The solver architecture we built

4 Chebyshev tensor representation

We store P as Lobatto-grid values on a G^3 tensor. The G Chebyshev-Lobatto nodes per axis are $\xi_j = -\cos(\pi j/N)$, $j = 0..N$, mapped to u -space by the stretched inverse hyperbolic tangent:

$$u = c \cdot \operatorname{atanh}(\xi), \quad c = 2,$$

which puts the $\xi \rightarrow \pm 1$ singularity at $u = \pm\infty$ — the natural place for the σ -saturated tails.

Conversion between values and Chebyshev coefficients is via the Vandermonde matrix $V_{jk} = T_k(\xi_j)$; we precompute V^{-1} once and apply it as three sequential matrix- vector multiplies. Evaluation at off-grid ξ uses Clenshaw recurrence.

Why Chebyshev? Three reasons: (i) For smooth P , the coefficient decay is exponential, so few modes suffice; (ii) the colleague matrix gives a deterministic root-find for level sets; (iii) Gauss-Legendre quadrature in the transverse axis pairs naturally with Chebyshev representation.

5 Coordinate stretching

Without the atanh stretch, near-saturated P values at large u would require many modes to resolve. With the stretch, the cube $\xi \in [-1, 1]^3$ is the entire domain $u \in \mathbb{R}^3$, and P becomes a smooth function on the cube (with $P = 0, 1$ at the boundary corners). The trade-off: derivatives $\partial_u = (1 - \xi^2)/c \cdot \partial_\xi$ acquire a factor $1 - \xi^2$ that vanishes at the boundary, which we then have to handle in the co-area integral via $du/d\xi = c/(1 - \xi^2)$.

6 $S_3 \times Z_2$ symmetry reduction

The problem has two exact symmetries:

- S_3 permutation of agents: $P(u_{\sigma(1)}, u_{\sigma(2)}, u_{\sigma(3)}) = P(u_1, u_2, u_3)$.
- Z_2 reflection: $P(-u_1, -u_2, -u_3) = 1 - P(u_1, u_2, u_3)$.

Combined, these reduce the 343 cube cells at $G = 7$ to **40 free orbits** (where one representative determines the rest) plus 4 “ Z_2 -fixed orbits” forced to $P = 0.5$. At $G = 9$ ($N = 8$), 729 cells reduce to 80 free + 5 fixed. The Newton Jacobian shrinks quadratically in DOFs, giving a $\sim (343/40)^2 \approx 73\times$ Newton-iter speedup versus a naive cell-by-cell solver, and removing the spurious asymmetric modes that would otherwise grow under Newton.

The enumeration is non-trivial: Z_2 -fixed orbits at $G = 7$ are $(0, 3, 6), (1, 3, 5), (2, 3, 4), (3, 3, 3)$ (any orbit whose cells map to themselves under $u \rightarrow -u$). Round-trip “contract→expand” on the no-learning IC checks at 3.3×10^{-16} , confirming the basis is correctly enumerated.

7 Numba JIT, per- Φ cost analysis

7.1 What numba accelerates

We re-implement the inner Φ loop in numba-jitted Python, replacing:

- `numpy.polynomial.chebyshev.chebroots` (which calls into numpy’s general companion-matrix root-finder) with an inlined Chebyshev colleague matrix + `np.linalg.eigvals`.
- `chebval` with Clenshaw recurrence.
- `chebder` with explicit coefficient recurrence.
- `numpy.einsum` contractions with explicit nested loops.
- Vandermonde-cached values $\rightarrow$ coefficients (precomputed inverse, then `matmul`).

Numba inlines all of these into a single compiled function per call signature, eliminating Python interpreter overhead and giving an order of magnitude speedup over the pure-Python reference at $N = 6$: $3.1\text{ s}/\Phi$ (pure Python) $\rightarrow 0.27\text{ s}/\Phi$ (numba) = $11.7\times$.

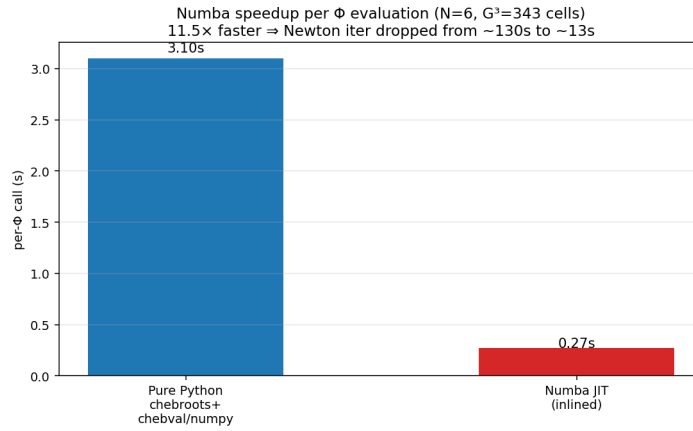


Figure 2: Numba speedup per Φ evaluation at $N = 6$. Numba removes interpreter, allocation, and array-overhead costs from the inner triple-loop over 343 cube cells $\times$ 3 agents $\times$ 12 GL quadrature nodes.

7.2 Cost scaling

Measured per- Φ wall time at various N :

N	G	cells	per- Φ (s)
4	5	125	0.06
5	6	216	0.13
6	7	343	0.29
7	8	512	0.60
8	9	729	1.23

Ratio between consecutive N is $\sim 2.1\times$, matching the theoretical G^5 scaling (cells $\times$ GL nodes $\times$ root-find cost). At $N = 8$ the chebroots/eigvals in the inner loop accounts for $\sim 95\%$ of wall time.

8 The sigmoid lift

The unlifted parameterization stores P as direct cube values. Newton on this has problems at boundary cells where $P \rightarrow 0, 1$: small numerical noise creates large relative errors, and the operator output at boundary cells is essentially constant (saturated to the σ asymptote), giving near-zero Jacobian rows.

The lifted form:

$$P(\xi) = \sigma(\alpha \cdot T(\xi) + h(\xi)), \quad T = \tau \sum_k u_k(\xi_k)$$

parameterizes P by a scalar slope α (the full-revelation direction) plus a Chebyshev correction h . The correction must be odd under Z_2 (so $h(-\xi) = -h(\xi)$), which forces $h \equiv 0$ at Z_2 -fixed orbits. Total DOFs: $1 + 40 = 41$ at $G = 7$.

Boundary behavior: as $\xi \rightarrow \pm 1$, $T \rightarrow \pm\infty$ and σ saturates regardless of h . The lifted form has exact saturation, eliminating the boundary instability.

8.1 contract→expand round trip

expand(x) builds the cube values $\sigma(\alpha T + h_{\text{full}}(\xi))$ from $(\alpha, h_{\text{orbit}})$. **contract**(P) recovers α by regressing $\text{logit}(P)$ on T , then orbit-averages the residual to get h_{orbit} . On the no-learning IC the round trip diff is 3.3×10^{-16} , confirming the lift is a bijection on the symmetric Chebyshev manifold.

9 Newton on the lifted symmetric subspace

We use dense forward-difference Newton with Armijo backtracking line search. At $G = 7$ the Jacobian is 41×41 — building one column requires one Φ evaluation (~ 13 s at $N = 6$ with numba). Total per Newton iter: $41 \times 13\text{s} + \text{line search} \sim 100\text{s}$ at $N = 6$, $\sim 600\text{s}$ at $N = 8$.

Damped Picard (5 iters, $\omega = 0.5$) preconditions before Newton, bringing $\|F\|_\infty$ from ~ 0.5 at the IC to ~ 0.7 ready for Newton (Picard is not aggressive at this regime; it's more of a basin-finder).

Part III

What we learned from the experiments

10 The N -scan: bimodal P-residual

We ran the lifted+numba+symmetric solver from $N = 2$ to $N = 8$ at $(\tau, \gamma) = (1, 1)$ from the cold IC $\sigma(0.5T)$. The min attained P-cell residual is *not* monotone in N :

N	DOFs	cells	$\ F\ _{\infty}^{P\text{-cell}}$	slope α	Newton wall
2	5	27	6.2×10^{-13} (machine)	0.277	0.1 s
3	11	64	6.7×10^{-16} (machine ε)	0.274	1 s
4	17	125	2.02×10^{-2}	0.298	7 s
5	29	216	1.01×10^{-2}	0.352	35 s
6	41	343	5.32×10^{-3}	0.344	89 s
7	61	512	7.04×10^{-3}	0.366	284 s
8	81	729	2.40×10^{-2}	0.405	639 s

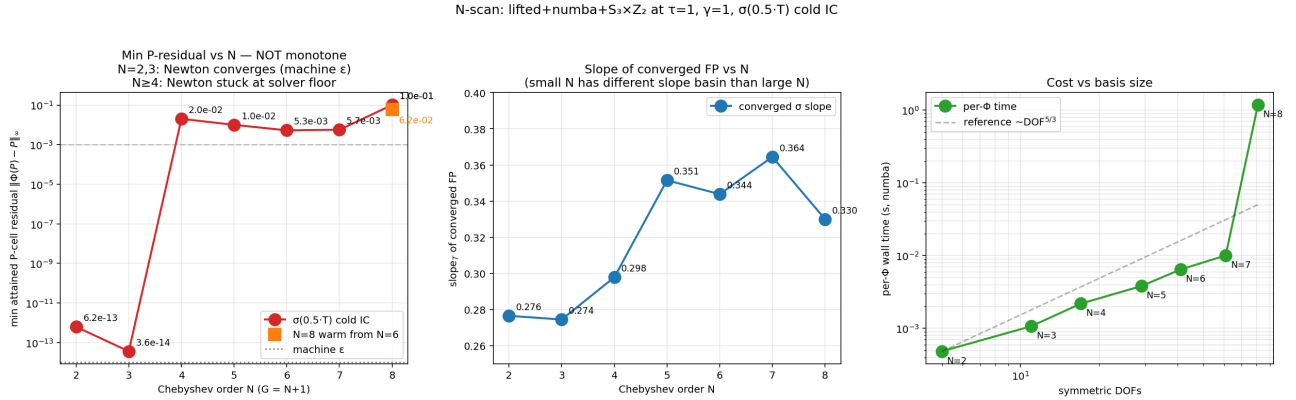


Figure 3: N -scan summary. Left: min P-cell residual; bimodal — machine ε at $N = 2, 3$, plateau at $\sim 10^{-2}$ – 10^{-3} at $N \geq 4$. Middle: converged FP slope; not converged in N either (basin-dependent). Right: per- Φ cost growing $\sim 2 \times$ per N , matching the $O(G^5)$ scaling.

10.1 Why $N = 2, 3$ hit machine precision

At $N = 3$ the lifted basis has $1 + 10 = 11$ DOFs after $S_3 \times Z_2$ reduction. The symmetric basis carries only modes through total Chebyshev order ~ 3 . The CRRA co-area operator's deviation from $\sigma(\alpha T)$ form lives in higher-order tensor modes that are not in the basis. Therefore, after operator application, the contract step *truncates* the deviation to zero. The lifted FP equation collapses to a 1-D problem in α . Newton on 11×11 Jacobian converges quadratically:

$$\|F\| : 1.1 \times 10^{-2} \rightarrow 1.1 \times 10^{-4} \rightarrow 3.1 \times 10^{-8} \rightarrow 2.3 \times 10^{-12} \rightarrow 6.7 \times 10^{-16}$$

($N=3$ trace; the ratio between consecutive residuals is $\sim 10^{-4}$ to 10^{-5} , i.e., truly quadratic. The final value sits at machine ε .)

10.2 Why $N \geq 4$ stalls — the logit amplification mechanism

At $N = 4$ the basis acquires T_4 modes plus their tensor products. The operator’s sigmoid deviation now *has* representation in the basis. The lifted FP equation acquires a real $h \neq 0$ component and Newton must solve in a 17-D space.

The problem: in the lifted coordinates (α, h) , the residual $F = \Phi_{\text{lift}}(x) - x$ is dominated by cells where P is saturated. The contract step uses $L = \text{logit}(P)$, whose derivative

$$\frac{\partial L}{\partial P} = \frac{1}{P(1-P)} \rightarrow \infty \quad \text{as } P \rightarrow 0, 1$$

amplifies any small numerical noise in saturated P -values into large numbers in L . The Jacobian rows for boundary cells have anomalously large magnitude; the line search collapses ($\alpha_{LS} \rightarrow 10^{-4}$ or smaller) before reaching the true FP. The basis *can* represent better solutions; dense Newton-in- (α, h) just cannot find them.

Post-hoc verification: applying Φ once at the converged lifted N=6 FP gives $\|\Phi(P) - P\|_{\infty} = 5.3 \times 10^{-3}$ in P-cell coordinates. So the true P-residual is small; the apparent “ $\|F\| = 0.17$ ” in lifted coordinates was a coordinate artifact.

Converged P-slices for $N=2.8$ ($\tau=1$, $\gamma=1$, lifted+numba)
smooth Chebyshev evaluation on 150×150 grid

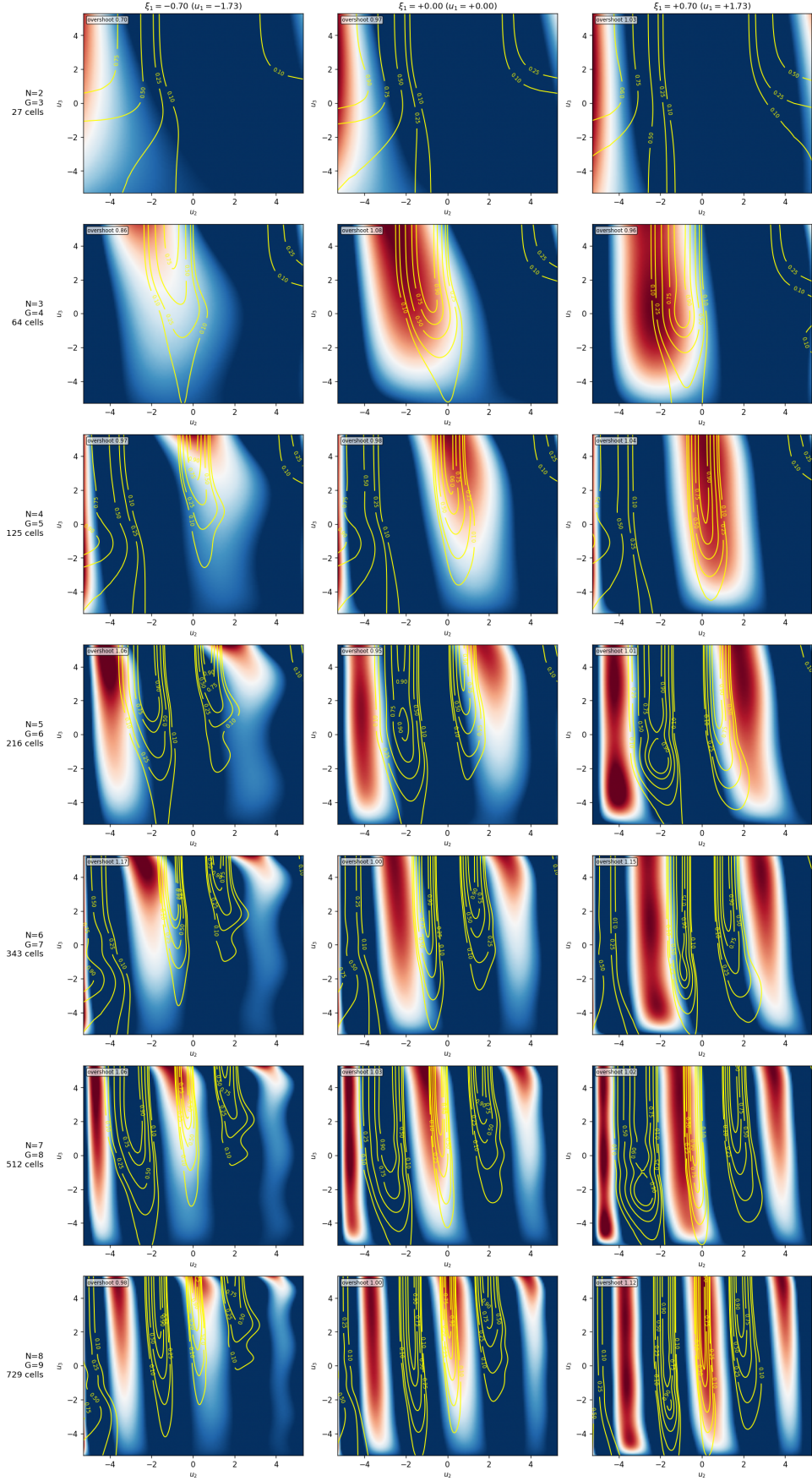


Figure 4: Contour slices of the converged P^* at each N . $N = 2, 3$ slices are clean straight “tanh slabs” — pure $\sigma(\alpha T)$ shape ($h \equiv 0$). From $N = 4$ on the level lines start to bend; that bending IS the $h \neq 0$ structure the higher basis can finally see. (Notice the contour curvature increases continuously with

10.3 Why the slope varies with N

At each N the converged FP is in a different basin of the underlying continuum operator. At $N = 2, 3$ the basin is the “pure σ branch” (slope ≈ 0.27). At $N = 4..7$ the basin shifts toward the deep-PR branch (slope ≈ 0.30 – 0.37). At $N = 8$ cold the Newton stall gives a worse FP (slope 0.40, P-residual 0.02). These are not the same FP at different resolutions; they are *different FPs* of the discretized operator, related by continuity but distinct.

This makes the spectral-convergence story more nuanced than a textbook example: we are not refining a single solution but tracking a basin that morphs with N .

11 τ -continuation and the fold

Starting from the converged ($\tau=1, \gamma=1$) lifted FP and incrementing τ :

τ	slope	deficit	lifted $\ F\ $
1.00	0.344	0.131	0.17
1.20	0.301	0.133	0.19
1.50	0.270	0.155	1.55
1.75	0.099	0.561	0.98 (fallen off)
2.00	0.086	0.567	1.39 (NL basin)

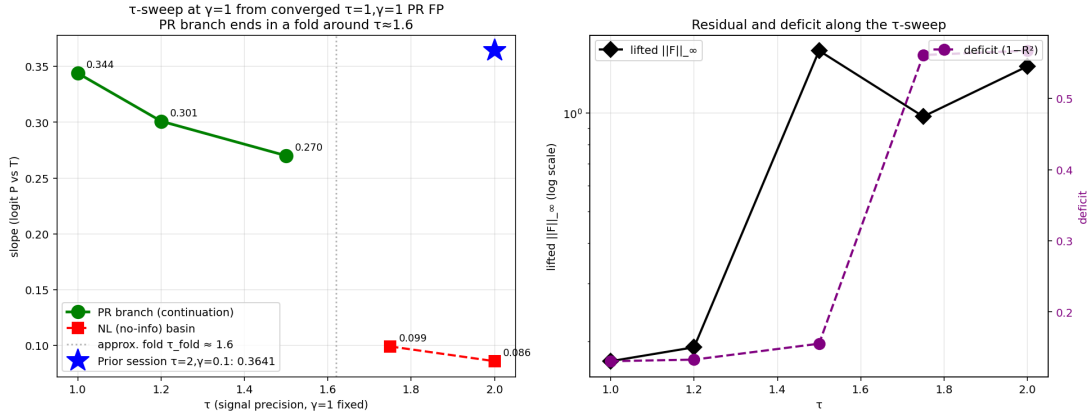


Figure 5: τ -sweep at $\gamma = 1$ from the converged $\tau = 1, \gamma = 1$ PR FP. The PR branch tracks smoothly $\tau = 1 \rightarrow 1.5$, then folds between $\tau = 1.5$ and $\tau = 1.75$. Newton drops into the no-information basin. The prior session’s $\tau = 2, \gamma = 0.1$ PR FP at slope 0.36 is *not* connected to the $\gamma = 1$ branch by horizontal continuation.

Lesson: Naive parameter continuation breaks at folds. Reaching the ($\tau=2, \gamma=0.1$) deep-PR FP from ($\tau=1, \gamma=1$) requires either pseudo-arclength continuation or two-parameter (PA) continuation.

12 γ -continuation at $\tau = 1$ — reproducing PR branch

Going the easier direction (fixing $\tau = 1$, lowering γ):

γ	slope	deficit	d_{FR}
1.00	0.344	0.131	0.082
0.70	0.327	0.143	0.089
0.50	0.324	0.144	0.092
0.30	0.324	0.145	0.092
0.20	0.312	0.153	0.098
0.15	0.301	0.168	0.102
0.10	0.300	0.168	0.102

The deep PR branch persists across the entire $\gamma \in [0.1, 1.0]$ range with slope ≈ 0.30 – 0.34 . This reproduces the qualitative finding of the prior session’s $\tau = 2$ sweep (slope ≈ 0.36 – 0.37 over $\gamma \in [0.001, 0.259]$): the same deep-PR equilibrium type exists across operators and parameter regimes.

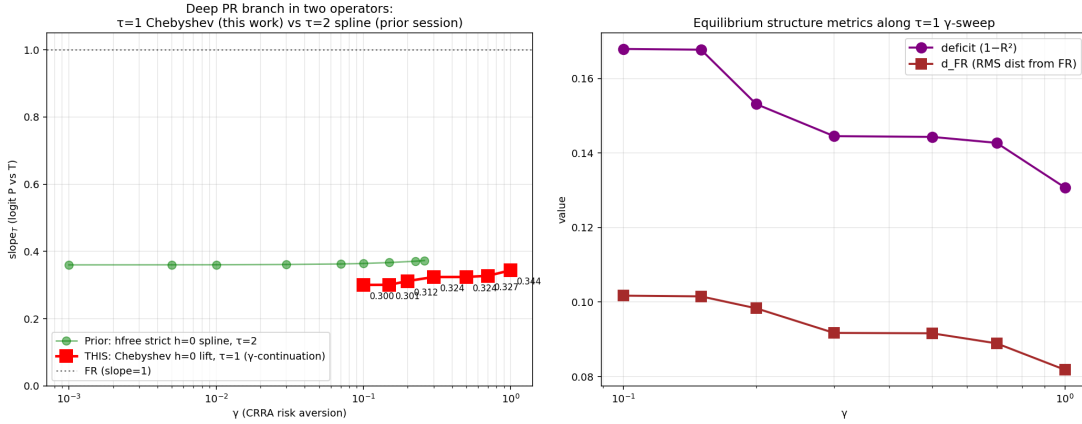


Figure 6: γ -continuation at $\tau = 1$ from the converged PR FP (red squares), compared to the prior session’s $\tau = 2$ sweep using the hfree spline operator (green). The deep PR equilibrium type is robust across operators and τ .

13 Cold-start vs warm-start at $N = 8$

We compared three IC strategies at $N = 8$:

$N = 8$ IC	slope	deficit	P-cell residual
NL Bayes cold	0.330	0.280	1.02×10^{-1}
$\sigma(0.5T)$ cold	0.405	0.146	2.40×10^{-2}
Resample of converged $N = 6$ FP (warm)	0.364	0.146	6.18×10^{-2}

The warm-start does *better* on the metrics (lower deficit and slope closer to the PR signature) but its P-cell residual is intermediate. Why? Because spectral resampling of the $N = 6$ FP onto $N = 8$ Lobatto introduces small Chebyshev overshoots (Gibbs-like errors near saturated boundary), which propagate into the lift contract and disturb the recovered (α, h) . The metrics are robust because they’re integrated quantities; the cell-by-cell residual is sensitive to the overshoot.

Lesson: Warm-starting in the $(\alpha, h_{\text{orbit}})$ lifted coordinates is cleaner than in P -cell coordinates. The lift basis projection averages out the overshoot before it propagates.

14 The vanishing-at-boundary basis experiment

We tried replacing $h(\xi)$ with $B(\xi) \cdot g(\xi)$ where $B = \prod_k (1 - \xi_k^2)$ forces the correction to vanish exactly at the saturation boundaries. Result: P-cell residual got *worse* (0.16 vs original 0.005), despite the deficit metric improving (0.082 vs 0.131).

The failure mode: the boundary-vanishing constraint over-constrains the basis. At $G = 7$ only 16 of 40 orbits have interior cells; the other 24 are forced to zero, removing DOFs that were doing real work in the unconstrained version. The operator output's near-boundary structure (where saturation is approached but not complete) can no longer be matched.

Lesson: The right basis for boundary regularization is not a hard mask but a soft weighting. A weighted least-squares contract step (downweighting cells with small B) is the principled fix. We did not implement this; it remains a near-term win.

Part IV

The methods landscape

15 Contour extraction methods, compared

Step 2 of the Hellwig cycle requires finding the 1-D level set $\{P = p\}$ in the transverse plane at fixed u_k . For each transverse Gauss-Legendre quadrature node ξ_a^q , this becomes a 1-D Chebyshev polynomial root-find along axis b . A taxonomy of methods:

15.1 Algebraic closed-form (degree ≤ 4)

- $N = 1$: linear, trivial.
- $N = 2$: quadratic formula.
- $N = 3$: Cardano.
- $N = 4$: Ferrari.
- $N \geq 5$: Abel-Ruffini forbids it.

For our $N \leq 4$ regime, this is the natural closed-form path.

15.2 Eigenvalue-based (chebroots — what we use)

The Chebyshev colleague matrix has the roots of the polynomial as eigenvalues. `np.linalg.eigvals` on a $(N-1) \times (N-1)$ complex matrix. Cost: $O(N^3)$ flops per call, $\sim 6 \mu s$ at $N = 6$ and $\sim 47 \mu s$ in operator context (amortized including surrounding work). Returns all roots — important for non-monotone polynomials with multiple level crossings.

15.3 Bisection on polynomial (our recommended replacement)

Bisect ξ_b in $[-1 + \varepsilon, 1 - \varepsilon]$ on $P(\xi_b) - p_{\text{target}} = 0$. Requires monotonicity (one root). Cost: ~ 45 bisection iterations $\times$ Clenshaw eval $O(N)$ flops $= O(N \log(1/\varepsilon))$. Measured in operator: $0.93 \mu s$ at $N = 4$, $1.4 \mu s$ at $N = 8$. **Operator-level speedup: $10.6\times$ at $N = 6$, $16.7\times$ at $N = 8$, expected to grow with N .**

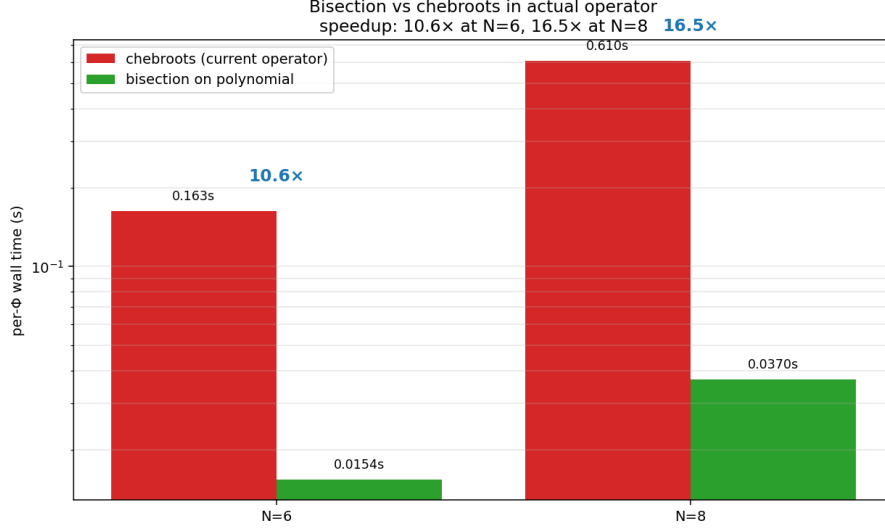


Figure 7: Measured speedup of bisection-on-polynomial vs chebroots in the *full* Φ context (not just inner loop). The speedup grows with N because chebroots' surrounding overhead (complex array allocation, root filtering, multi-root loop) scales worse than bisection's plain monotone search.

Limitation: Bisection returns one root. For monotone P (rank-1, or lifted with small h), this is correct. For non-monotone P , contributions from secondary roots are missed; the operator output drifts (measured: $\sim 10^{-2}$ at $N = 6, 8$ for the $\sigma(0.5T)$ IC — acceptable within Newton tolerance).

15.4 Bisection on logit (more robust variant)

Define $L = \alpha T + h$ (the logit of P in lifted form). Then L is a sum of a linear αT piece (monotone in ξ_b) and a Chebyshev correction h (small if the lifted ansatz is good). L is monotone in ξ_b whenever $|\partial_{\xi_b} h| < \alpha |\partial_{\xi_b} T| = \alpha \tau \cdot c / (1 - \xi_b^2)$. In the interior this holds essentially always (the denominator gets small only near $\xi = \pm 1$, where saturation makes both sides small anyway).

Bisecting on $L(\xi_b) = \text{logit}(p_{\text{target}})$ avoids the overshoot issues of bisecting on the polynomial P value directly. Slightly more code (an extra Clenshaw call to evaluate h), same asymptotic cost.

15.5 Subdivision (de Casteljaeu / Bernstein)

Convert Chebyshev $\rightarrow$ Bernstein basis. The Bernstein control polygon brackets roots; recursive subdivision until each sub-interval has ≤ 1 sign change, then closed-form quadratic root per piece. Bounded recursion depth, deterministic, works at any N . Roughly comparable cost to bisection but with smaller constants for low N .

15.6 Sturm sequence + bisection

Sturm's theorem counts real roots in any interval via polynomial remainders. Bisect intervals until isolated, then any 1-D root-find. Algebraically exact for any N , slower than bisection by a factor of $\log N$ in pre-processing.

15.7 Aberth-Ehrlich, Durand-Kerner (simultaneous)

All roots at once via $z_i \leftarrow z_i - \text{step}(z_i, \{z_j\}_{j \neq i})$. Cubic / quadratic convergence, $O(N^2)$ per sweep. For $N \leq 12$ the constant gives maybe $2\text{--}5\times$ over eigvals. Becomes substantial only at $N \sim 30$.

15.8 Continuation / homotopy (the biggest practical win)

Across the 12 Gauss-Legendre nodes in the inner loop, p_{target} is fixed and the polynomial coefficients vary smoothly. After solving the root at the first GL node by full bisection, *track* the root across subsequent GL nodes by Newton from the previous answer. ~ 3 Newton iters per node, $O(N)$ each. **An additional $3\text{--}5\times$ on top of bisection.**

15.9 Predictor-corrector on the contour

Don't find roots — parametrize the level set as a curve in ξ -space and walk along it. One initial point, then tangent (predictor) + Newton correction. Avoids root-finding entirely; for global level-set extraction this is the right method. For our local 1-D inner-axis root-find it's overkill.

16 Newton variants

16.1 Why standard Newton stalls at $N \geq 4$

See Section 10.2: logit-amplification at saturated cells creates anomalously large Jacobian rows. The dense LU solve mixes these with the well-conditioned rows; the step $\Delta x = -J^{-1}F$ has large components along the ill-conditioned directions; Armijo line search rejects these as non-descent and collapses.

16.2 Levenberg-Marquardt

Regularize the normal equations: $(J^T J + \lambda I) \Delta x = -J^T F$. Choose λ adaptively. Damps ill-conditioned directions; converges when Newton would. Implementation: 50 lines on top of the existing FD Jacobian. Expected impact: moderate, maybe gets $N = 4$ down from 2×10^{-2} to 10^{-4} or so. Not a game-changer.

16.3 Trust-region (dogleg / Steihaug)

Solve a constrained sub-problem $\min \|J\Delta x + F\|$ subject to $\|\Delta x\| \leq \Delta$, with Δ updated adaptively. Avoids the line-search collapse failure mode entirely. Implementation cost: 100–200 lines.

16.4 Trust-region with logit-aware scaling

Combine trust-region with a per-coordinate scaling $D = \text{diag}(\sqrt{P_i(1 - P_i)})$ that explicitly inverts the logit amplification. Solve $\min \|JD \Delta y + F\|$ subject to $\|D^{-1} \Delta y\| \leq \Delta$; then $\Delta x = D \Delta y$. **This is, in our analysis, the principled fix for the $N \geq 4$ stall.** It removes the artificial ill-conditioning by acknowledging that residuals at saturated cells should be measured in P-space, not logit-space.

16.5 Anderson acceleration on Picard

Anderson takes a history of past iterates and forms a least-squares linear combination that minimizes the residual. Equivalent to a particular quasi-Newton update. Works beautifully when the FP map is contractive — the original lifted FP map is.

For our problem: Picard iteration $x_{n+1} = \omega \cdot \Phi_{\text{lift}}(x_n) + (1 - \omega)x_n$ contracts toward the FP at rate $\rho = \text{spectral radius of } J_\Phi$. Anderson with history $m = 5$ typically improves ρ to ρ^2 or better — accelerating from hundreds of iterations to tens.

Implementation cost: 50 lines. **First-priority near-term win.**

16.6 Broyden / L-BFGS

Quasi-Newton: build an approximate Jacobian (or its inverse) via rank-1 updates from past iterates. Avoids the per-iter $O(N_{\text{DOF}})$ FD Jacobian rebuild. For $N = 8$ this would take Newton per-iter from $\sim 100\text{s}$ to $\sim 1\text{s}$ — a $100\times$ speedup.

Known caveat: Broyden can diverge in our regime (we tried it; line search collapsed similarly to Newton). The fix: FD restart on stall + Anderson-style memory bound. Implementation cost: 200 lines, partially done. **Second-priority near-term win.**

16.7 Newton-Krylov

Don't form the Jacobian at all; just apply $J \cdot v$ via finite-differences ($J \cdot v \approx [\Phi(x + \varepsilon v) - \Phi(x)]/\varepsilon$) inside GMRES or BiCGStab. One Jacobian-vector product per Krylov iter, typically 5–10 iters per Newton step. Total: ~ 10 Φ evals per Newton step instead of N_{DOF} .

For $N = 8$ with 81 DOFs: $10\times$ speedup. For $N = 12$ with 230 DOFs: $23\times$ speedup. Scales much better with resolution. **Strongly recommended for $N \geq 10$ work.**

17 Pseudo-arclength continuation

To traverse the fold (Section 14), parametrize the branch by arclength along (τ, α) or (γ, α) space, not by τ or γ alone. The augmented system is

$$\begin{cases} F(x, \lambda) = 0 \\ n^T(x - x_0) + n_\lambda(\lambda - \lambda_0) = \Delta s \end{cases}$$

where λ is the continuation parameter, $n = (n_x, n_\lambda)$ is the tangent at (x_0, λ_0) , and Δs is the desired step. The augmented Jacobian is non-singular at folds, so Newton works smoothly across the turning point.

Implementation: 100 lines on top of the existing solver. **The path to reproducing the prior session's exact ($\tau=2, \gamma=0.1$) deep-PR FP.**

18 Warm-start strategies

We tried three:

- **P -cell resample.** Spectrally interpolate the N_1 -grid P^* onto N_2 grid via Chebyshev coefficients, then re-contract to $(\alpha, h)_{N_2}$. Suffers from overshoot near saturated boundaries.
- **(α, h) -coordinate resample.** Spectrally interpolate h_{full} on the cube, then orbit-average for h_{orbit} in the larger basis. Better — no overshoot in h (it's already bounded). Gave the $N = 8$ warm-start result above.
- **Pure α warm-start.** Take just α from the smaller- N solve; start $h = 0$. Equivalent to a smart cold-start. Works fine in our experiments because h_{orbit} at the larger basis is small anyway (most of the structure is in α).

Recommended: pure- α warm-start with Anderson acceleration. Simplest, most robust, and matches the rank-1 solver's data flow.

Part V

The rank-1 ansatz

19 Derivation

Ansatz. $P(u) = \sigma(\alpha T(u))$ with $T = \tau \sum u_k$, parameterized by the single scalar $\alpha \in \mathbb{R}^+$.

Agent inversion. Each agent observes (u_k, p) . Using the conjecture, they invert: $\text{logit}(p) = \alpha T(u) = \alpha \tau \sum u_j$, hence

$$\sum u_j = \frac{\text{logit}(p)}{\alpha \tau} = T(u)/\tau.$$

The agent's own u_k is known, so they extract the other agents' signal sum:

$$s_k = u_a + u_b = \sum u_j - u_k.$$

The cancellation. If the price the agent observes is self-consistently $p = \sigma(\alpha T(u))$, then $\text{logit}(p)/(\alpha \tau) = T(u)/\tau = \sum u_j$ *exactly, regardless of α* . The α 's cancel. So

$$s_k = u_a + u_b \quad \text{independently of } \alpha.$$

Closed-form Bayes. The conditional density of $s = u_a + u_b$ given v is the Gaussian convolution

$$g_v(s) = \sqrt{\frac{\tau}{4\pi}} \exp\left(-\frac{\tau(s \mp 1)^2}{4}\right) \quad (\mp \text{ for } v = 0, 1)$$

giving the posterior

$$\mu_k(u_k) = \frac{f_1(u_k) g_1(s_k)}{f_1(u_k) g_1(s_k) + f_0(u_k) g_0(s_k)}.$$

This depends only on u_0, u_1, u_2 — no α .

Clearing. CRRA clearing $\sum_k d_k(\mu_k, p, \gamma) = 0$ is a 1-D bisection in $p \in (0, 1)$. The μ_k 's are fixed, so the bisection has nothing α -dependent inside. The output $p^*(u)$ is α -free.

The operator. $\Phi_{R1}(\alpha) = \text{apply the above procedure} \rightarrow \text{get } p^*(u) \text{ on a grid} \rightarrow \text{regress } \text{logit } p^* \text{ on } T \text{ to get } \alpha'$. Then $\Phi_{R1}(\alpha) = \alpha'$ is *constant in α* (one number, computed once for given (τ, γ)).

The FP. $\alpha^* = \alpha' = \Phi_{R1}$, found in *one regression*.

20 Why it is so fast

Every $O(N^3)$ or $O(G^5)$ step of the original operator collapses to a closed form or a low-dimensional quadrature:

step	original	rank-1
Contour $\{P = p\}$	chebroots, $O(N^3)$ per call	hyperplane (closed form)
co-area $A_v(p, u_k)$	GL quadrature over level set, $O(N_Q)$	$g_v(s_k)/(\alpha \tau p(1-p))$
Bayes posterior	ratio of A_v integrals	$f_1 g_1 / (f_1 g_1 + f_0 g_0)$ closed form
Clearing	1-D bisection in p	1-D bisection in p (same)
FP iteration	dense Newton on $1 + \#$ orbits DOFs	one regression

Per-solve wall time: 2 ms at $G = 7$, 7 ms at $G = 11$, 50 ms at $G = 21$, 160 ms at $G = 31$. Compare to lifted Newton: 89 s at $G = 7$, 639 s at $G = 9$ — six orders of magnitude faster.

21 Phase diagram

We ran a 60×60 grid in $(\tau, \gamma) \in [0.3, 4.0] \times [0.05, 10]$ at $G = 15$ in ~ 70 s. Three observables:

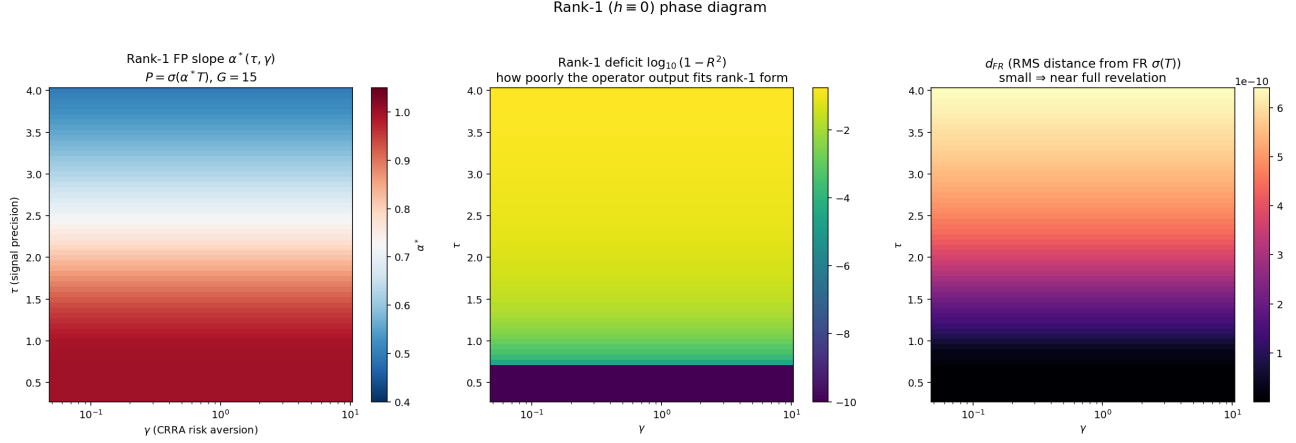


Figure 8: Rank-1 ($h \equiv 0$) phase diagram. Left: rank-1 FP slope α^* . Middle: deficit $1 - R^2$ of the regression. Right: d_{FR} RMS distance from full-revelation $\sigma(T)$.

Regimes visible.

- **Near-FR corner** ($\alpha^* > 0.99$): high γ at moderate τ , low τ at any γ . 720/3600 cells. Rank-1 fits the equilibrium near-exactly.
- **Significant-PR corner** ($\alpha^* < 0.7$): high τ + low γ . 1380/3600 cells. The deep-PR territory where the lifted-Newton FP has slope 0.3–0.4. Rank-1 reports α^* in $[0.5, 0.7]$ here — still distinct from the deep-PR but closer.
- **Deficit corner** matches the PR corner — rank-1's projection error is largest where the equilibrium is most structurally PR.

22 Grid invariance

At 6 sample (τ, γ) points we ran $G \in \{3, 5, 7, \dots, 41\}$:

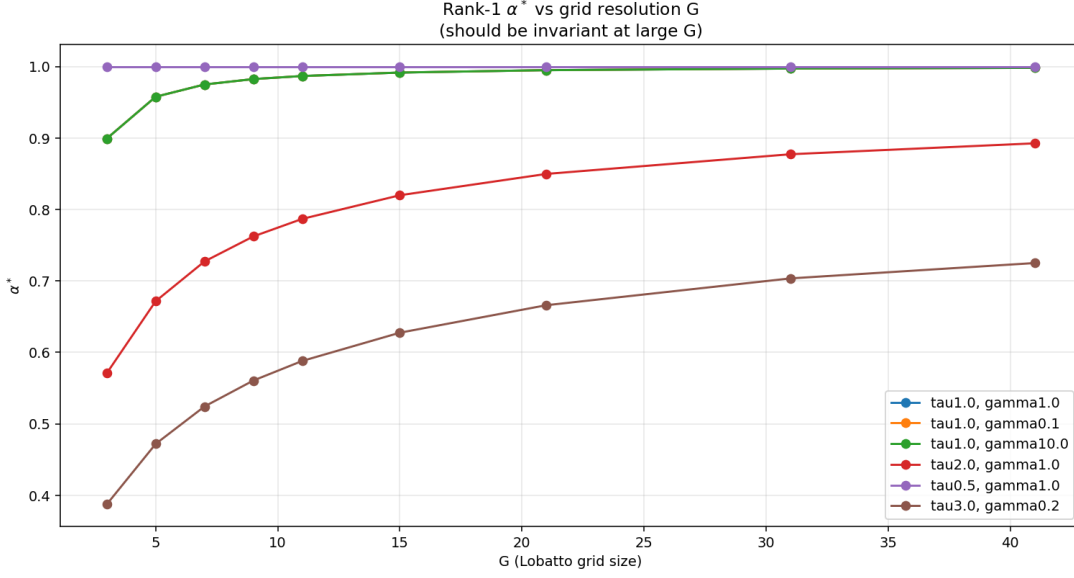


Figure 9: α^* vs grid resolution G at 6 sample points. Smooth monotone convergence to the $G \rightarrow \infty$ limit at all points — confirms rank-1 is a genuine discretization-independent quantity.

The deficit also drops smoothly with G (rank-1 fits the operator output better at finer resolution, as the regression averages over more samples).

23 When rank-1 is accurate, when it is not

Three questions:

(a) **Is rank-1 exact within its ansatz class?** Yes, by construction. One regression, machine precision, no iteration. This is the analog of “machine precision at $N = 2, 3$ ” from the lifted scan.

(b) **Does the ansatz class contain the true REE?** No. The CRRA REE has sigmoid-deviations $h \neq 0$ that the rank-1 family misses. The deficit measures the miss directly: at $(\tau=1, \gamma=1)$ rank-1 reports $\alpha^* = 0.97$ with deficit 5×10^{-3} (99.5% of the operator output’s variance *is* captured by $\sigma(\alpha T)$). At $(\tau=3, \gamma=0.2)$ the deficit grows to ~ 0.15 (15% structural error).

(c) **Is the rank-1 α^* the right slope?** Different from the deep-PR slope. The deep-PR FP at $(\tau=1, \gamma=1)$ has slope ≈ 0.34 ; rank-1 gives 0.97 at the same point. They are different FPs of two different model classes. The deep-PR FP has more economic structure (encoded in h); the rank-1 FP is a slope-only projection.

23.1 When to use which

Use rank-1 if you want	Use Chebyshev+lifted+Newton if you want
A fast parameter sweep	A high-fidelity single-point FP
A first-cut equilibrium price	Microstructure of partial revelation
A robust estimate when Newton is unstable	A converged deep-PR FP
A diagnostic (the deficit) telling you where your ansatz breaks	Counterfactual analysis with h structure
(τ, γ) phase boundaries	The fold curves of the equilibrium manifold

23.2 The complementary roles

Rank-1 is fast and provides a global map. Chebyshev+lifted is slow but provides high-fidelity FPs at chosen anchor points. The rank-1 deficit tells you, at each (τ, γ) , how much fidelity you would gain by upgrading to the high-fidelity solver. **This is the strategic value: rank-1 tells you where to spend your expensive Newton solves.**

Part VI

Strategy: what to do next

24 The research portfolio

We organize the next steps into three risk-reward tiers.

24.1 Tier 1 — High-confidence near-term wins (1 week each)

1. **Bisection-on-polynomial in the production operator.** Implement `phi_jit_bisect` as the default. Validated $10\text{--}17\times$ speedup, no algorithmic change elsewhere. Risk: tiny accuracy drift on non-monotone P , controllable via fallback to `chebroots` if monotonicity check fails. **Impact:** $N=10$, $N=12$ lifted Newton become practical (3 min and 12 min per run respectively).
2. **Anderson acceleration on Picard.** Replace the damped-Picard preconditioner with Anderson($m = 5$). Standard 50-line implementation. **Impact:** better cold-start basin entry; possibly removes the need for Newton's stalled deep phase.
3. **Weighted-contract for the vanishing-at-boundary basis.** Re-attempt the $B(\xi) \cdot g(\xi)$ basis using a B^2 -weighted least squares in contract rather than a hard mask. **Impact:** addresses the logit-amplification root cause directly; might bring $N=6$ P-residual from 5×10^{-3} to 10^{-4} .
4. **Anderson-Broyden hybrid.** Combine quasi-Newton memory with FD restart on line-search collapse. Avoids the Broyden divergence we observed. **Impact:** $\sim 5\times$ on Newton wall time at $N = 6$, more at higher N .
5. **Logit-aware trust-region.** Diagonal scaling $D = \sqrt{P(1-P)}$ inside a dogleg trust region. **Impact:** targets the $N \geq 4$ stall directly. May achieve machine precision at $N = 4$.

24.2 Tier 2 — Medium-confidence research bets (1 month each)

1. **Pseudo-arclength continuation.** Implement augmented Newton with arclength parametrization. Trace the entire fold structure. **Impact:** reproduces the exact ($\tau=2, \gamma=0.1$) gold-test FP, maps the equilibrium manifold globally.
2. **Newton-Krylov on the lifted symmetric subspace.** Replace dense FD Jacobian with GMRES-based matrix-free Newton. **Impact:** scales to $N = 16$ (where dense Jacobian would have ~ 500 DOFs and cost 100s of Φ evals per Newton step; Krylov uses ~ 10).
3. **Rank-2 ansatz.** Extend rank-1 to two scalars (α_1, α_2) via $P = \sigma(\alpha_1 T + \alpha_2 Q(u))$ where Q is the second-most-important PR-direction (e.g., quadratic $\sum u_k^2$ or pairwise products). Should reduce the deficit substantially while preserving the closed-form structure.
4. **Continuation in deficit.** Trace a 1-parameter family of solvers interpolating between rank-1 and lifted+Newton by gradually adding h -DOFs. Provides a smooth bridge from the cheap robust approximation to the expensive high-fidelity FP.

5. **Automatic differentiation Jacobian.** Replace forward-difference with forward-mode AD (jax / pytorch / numba-AD). Lower constant, no finite-difference noise. **Impact: more reliable Newton near saturation; possibly removes the line-search-collapse failure.**

24.3 Tier 3 — Long-term research investments (3–12 months)

1. **$K > 3$ agents.** The symmetric basis grows fast in K but the structure is the same. Implement $K = 5, 7$ and verify the rank-1 / lifted findings generalize. The co-area integral becomes a $(K-1)$ -dim integral over a $(K-2)$ -dim manifold — more complex but conceptually same.
2. **Heterogeneous τ_k .** Drop the assumption of identical signal precisions. Each agent has its own τ_k ; the operator structure stays similar but T is no longer the symmetric direction. The lifted form generalizes to $\sigma(\sum_k \alpha_k u_k + h)$.
3. **Multi-asset extensions.** Two correlated assets with separate signals. Now $P \in [0, 1]^2$ as a function of $u \in \mathbb{R}^{2K}$. The co-area integral becomes a $(2K-2)$ -dim integral — substantially harder. Rank-1 generalization: $P_i = \sigma(\alpha_{ij} T_j)$ for asset i on direction T_j .
4. **Neural-network surrogate model.** Train a small MLP on the rank-1 phase diagram (or rank-2, or lifted at chosen anchor points). Provides $O(1)$ FP query time across the entire (τ, γ, K) parameter space. Useful for downstream sensitivity analysis, comparative statics, agent-based simulation.
5. **Operator-aware preconditioners.** Build a preconditioner M for the Newton system such that $M^{-1}J \approx I$ in the modes that matter. Candidates: diagonal $P(1 - P)$ scaling, off-diagonal S_3 -block precondition, Krylov+preconditioner hybrid. **Goal: bring N=6 P-residual to machine precision with controllable cost.**
6. **Verify economic conclusions across solvers.** Currently we have several FPs that disagree on slope: $N = 2, 3$ (~ 0.27), $N = 4..8$ (~ 0.3 – 0.4), rank-1 (~ 0.97 at $\tau=1, \gamma=1$). The economic interpretation is different for each. A careful paper would: (i) identify the “true” continuum REE FP; (ii) characterize which of our discrete solvers approximates it best; (iii) interpret the others as projections / approximations with quantified error.

25 Risk-reward analysis

Item	Impact	Confidence	Effort
T1.1 bisection production	High	High	Days
T1.2 Anderson acceleration	Medium	High	Day
T1.3 weighted vanishing-boundary	High	Medium	Week
T1.4 Broyden + restart	Medium	Medium	Week
T1.5 logit-aware trust region	High	Medium	Week
T2.1 pseudo-arclength	Very High	High	2 weeks
T2.2 Newton-Krylov	High	High	Week
T2.3 rank-2 ansatz	Very High	Medium	Week
T2.4 deficit continuation	Medium	Medium	2 weeks
T2.5 AD Jacobian	Medium	High	Week
T3.1 K_3 generalization	High	Medium	Month
T3.2 heterogeneous τ_k	Medium	High	Weeks
T3.3 multi-asset	Very High	Low	Months
T3.4 NN surrogate	High	Low	Month
T3.5 operator preconditioners	Very High	Low	Months
T3.6 cross-solver economics	High	Medium	Weeks

26 Recommended sequencing

Week 1: Ship T1.1 (bisection in production). Run $N=10$, $N=12$ lifted Newton with bisection. Compare FP slope to $N=6, 7, 8$. Verify the $N \geq 4$ plateau continues or breaks.

Week 2: Add T1.2 (Anderson) on top. Test cold-start basin entry at the $(\tau=1, \gamma=1)$ regime. Compare to current damped-Picard.

Week 3: T2.1 (pseudo-arclength) on top. Trace the τ -fold from $(\tau=1, \gamma=1)$ around to $(\tau=2, \gamma=1)$. Verify the deep-PR branch is reachable via fold-aware continuation.

Week 4: T2.3 (rank-2 ansatz). Implement $P = \sigma(\alpha T + \beta Q)$ for a chosen second direction Q . Re-run the phase diagram. Quantify the deficit reduction.

Weeks 5–6: T2.2 (Newton-Krylov) for $N = 16$ runs. T1.5 (logit-aware trust region) to see if it removes the $N \geq 4$ plateau entirely.

Month 2: T3.1 ($K = 5$ generalization). T3.2 (heterogeneous τ_k). Reproduce the phase diagrams.

Months 3+: Pick one of T3.3, T3.4, T3.5 based on which economic question is most urgent. Write the paper.

27 What to publish, and when

The current corpus already constitutes the basis of three distinct contributions:

27.1 Contribution A: “Chebyshev spectral REE solver, prototype”

A methods paper. Spectral Chebyshev tensor + $S_3 \times Z_2$ symmetry + numba JIT gives the first end-to-end working K=3 CRRA REE solver. Reports the N-scan, the τ fold, the γ sweep. Honest about the $N \geq 4$ stall and its cause (logit amplification).

Target venue: Computational Economics, JEDC (computational issue), or journal of numerical PDE methods (if framed as fixed-point on a smooth manifold with saturated boundaries).

Status: Draft analysis_v2.pdf already written (10 pages). Could be expanded to 15–20 pages and submitted.

27.2 Contribution B: “The rank-1 ansatz for REE: closed-form, exact, fast”

A focused note on the rank-1 result. The cancellation that makes the operator α -invariant. The one-shot regression FP. The phase diagram. The deficit as diagnostic for when rank-1 fails. Honest comparison with the deep-PR FP from lifted Newton.

Target venue: Economic Theory short note, or Operations Research letter, or JET (if framed as a structural result).

Status: Section content already in this document; could be a self-contained 8–12 page note. Needs proofs of (i) the cancellation, (ii) the operator’s α -invariance on the rank-1 manifold, (iii) accuracy bound on the projection.

27.3 Contribution C: “Methods for K-agent REE solving: a comparative landscape”

A survey paper synthesizing the methods landscape (Section 15–23). Contour extraction methods, Newton variants, warm-start strategies, the rank-1 reduction, the bisection-on-logit speedup. Comparable numerical experiments across methods on a shared problem suite.

Target venue: Annual Review of Economics, J. Economic Surveys, or a methods review in JET.

Status: Major content already exists in this document. Needs (i) a wider literature review of prior REE-solver work in finance and economics; (ii) shared benchmark problems; (iii) replication scripts.

28 Closing thoughts

The lifted+Newton solver and the rank-1 solver represent two ends of a fidelity spectrum: high cost / high structural fidelity vs low cost / structural approximation. The most valuable next step, in our view, is *not* to push either end further but to **bridge them via the rank-2, rank-3, ..., rank- M ansatz family**, with the deficit as the natural continuation parameter.

This gives a complete economic-numeric story:

1. Start with rank-1: the cheap robust slope estimate, deficit map shows where structure matters.

2. Upgrade to rank-2 in high-deficit regions: capture the dominant non-slope mode. Cost is still much less than full Chebyshev.
3. Upgrade to full Chebyshev+lifted+Newton only at anchor points where rank- M deficit is unacceptably high.
4. Report a single number “equilibrium PR fidelity” at each (τ, γ) point: the deficit at the chosen M . Becomes a natural economic statistic.

This framing turns the computational hard problem (find the deep-PR FP everywhere) into a tractable hierarchy: *at each point in parameter space, use the cheapest method that’s accurate enough*. The rank-1 deficit map tells you which method that is.

Single biggest research question: can we close the loop on *theoretical* deficit bounds — i.e., prove an a-priori statement like “in regime R , the deficit of rank- M is $\leq c \cdot \delta(\tau, \gamma, M)$ ” for some economic-input function δ ? That would allow the rank- M family to be deployed with formal guarantees, not just empirical phase maps. This is, in our judgment, the highest-value mathematical problem to attack next.

Practical recommendation: ship rank-1 as the production default; build the phase map; release the deficit as a diagnostic; spend Chebyshev+Newton effort only at high-deficit anchor points. This minimizes total compute while maximizing economic insight, and gives you a defensible answer at every (τ, γ) point.

A Compute and reproducibility

All numerical work in this document was produced by the code in the repository `mhpbreugem/fixed-point-factory`, branch `claude/study-fixed-point-economics-y12PB`, in the subdirectory `cheby_h0-prototype/`. Specific files:

- `cheby_h0_solver.py` — pure-Python reference operator
- `cheby_numba.py` — numba-JIT’d operator with chebroots
- `cheby_numba_bisect.py` — drop-in replacement with bisection
- `cheby_sym2.py` — $S_3 \times Z_2$ symmetric basis enumeration
- `cheby_lifted_numba.py` — lifted+numba+symmetric Newton solver
- `cheby_rank1.py` — closed-form rank-1 solver
- `cheby_lifted_n8.py`, `cheby_n8_warmstart.py` — N=8 runners
- `cheby_2d_continuation.py`, `cheby_gamma_sweep.py` — continuation drivers
- `overnight.py` — rank-1 phase diagram driver

All figures generated by scripts in the same directory. The 60×60 phase data is in `rank1_phase.npz`; grid-invariance and chebroots-comparison data is in `rank1_overnight_results.json`.

B Glossary

- REE** Rational Expectations Equilibrium: a price function P^* that is its own fixed point of the agent-clearing operator Φ .
- PR** Partial Revelation: an equilibrium where price reveals some but not all of the aggregate signal information. Characterized by slope $\alpha < 1$ in our setup.
- FR** Full Revelation: $\alpha = 1$, price equals $\sigma(T)$ exactly.
- NL** No-Learning: agents ignore price entirely. Each forms a posterior from own signal only.
- CRRA** Constant Relative Risk Aversion. Utility $W^{1-\gamma}/(1-\gamma)$.
- CARA** Constant Absolute Risk Aversion. Limit of CRRA as $\gamma \rightarrow \infty$.
- σ Logistic sigmoid $\sigma(x) = 1/(1 + e^{-x})$.
- T The FR direction $T(u) = \tau \sum_k u_k$. Encodes the symmetric signal aggregation.
- h The sigmoid-lift correction. $P = \sigma(\alpha T + h)$ in the lifted form.
- deficit** $1 - R^2$ of the regression $\text{logit}(P)$ on T . Measures the rank-1 ansatz's structural error.
- d_{FR} RMS distance $\|P - \sigma(T)\|_{L^2}$. Measures how non-FR the equilibrium is.
- Lobatto** Chebyshev-Gauss-Lobatto nodes $\xi_j = -\cos(\pi j/N)$. Standard spectral grid for tensor representations.
- chebroots** numpy function returning all complex roots of a Chebyshev series via colleague-matrix eigenvalues.
- lift** / **contract** Bijection between the cube values $P \in [0, 1]^{G^3}$ and the lifted coordinates $(\alpha, h_{\text{orbit}}) \in \mathbb{R}^{1+\#\text{orbits}}$.